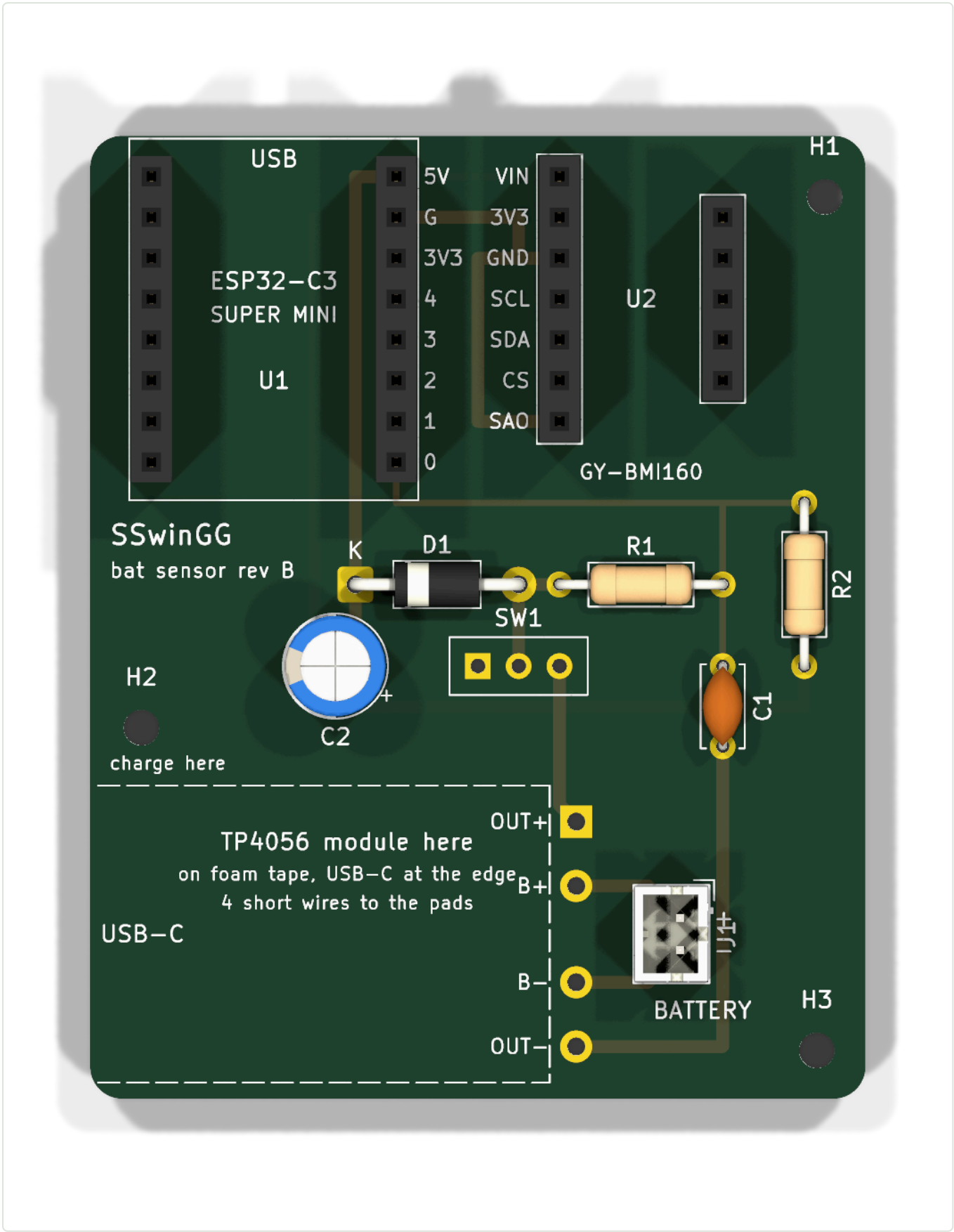
# The printed circuit board




The same circuit as the zero PCB, on a board made in a factory. The wires are copper tracks, so nothing is wired by hand and nothing can be wired wrong. Everything is on the one board. The ESP32 and the sensor

plug into sockets. The charger module sits in its marked place along the bottom, with its USB-C socket at the edge, joined by four short wires, so a module from any maker fits. The cell is taped to the back in its marked place and plugs into a JST socket. The board is 48.3 by 60 mm with two copper layers.

The design is a KiCad 10 project in pcb/: a schematic and a board layout, with the parts of this kit as a small library of their own. KiCad is free, from kicad.org.

## What has been checked, and what has not

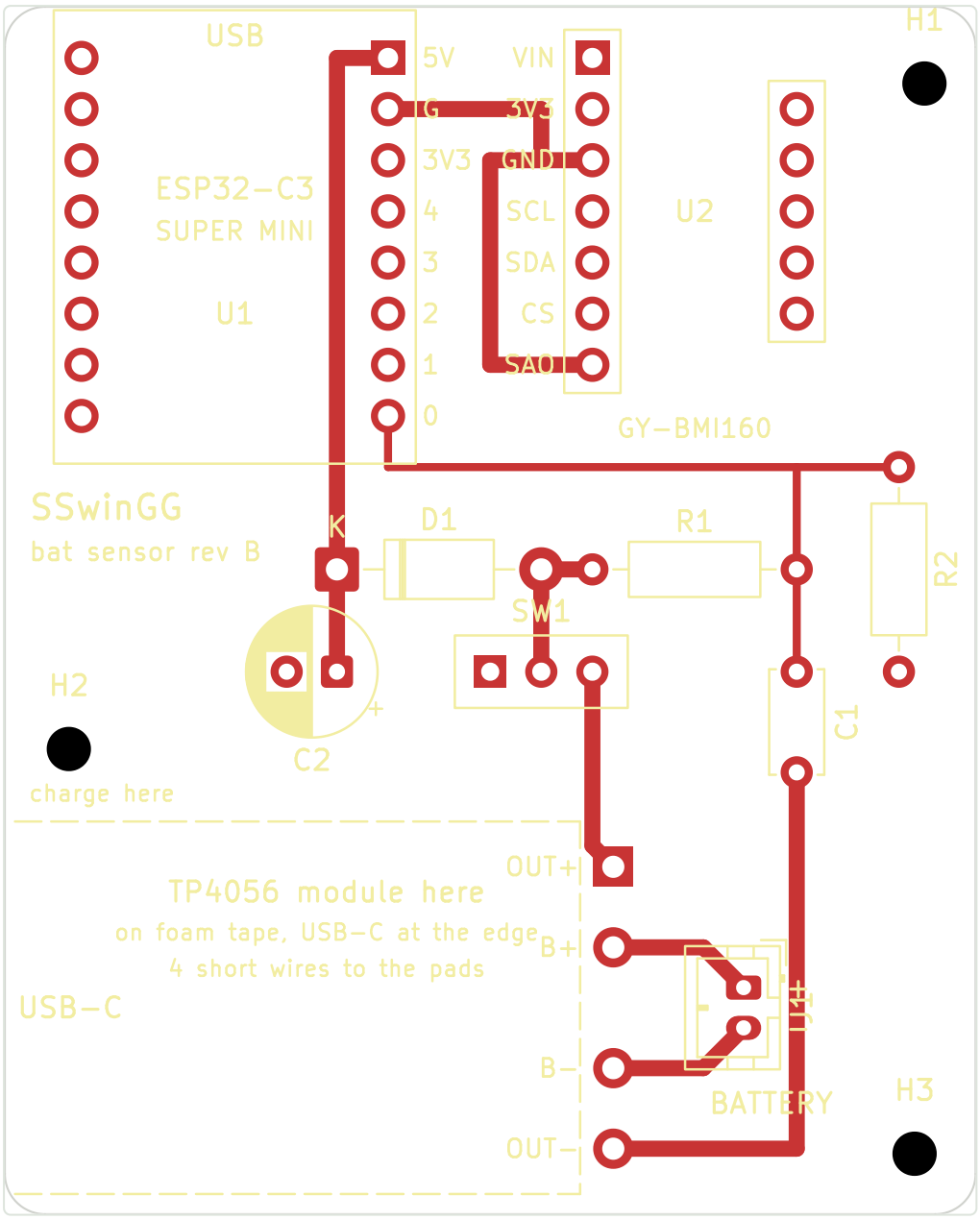
- **Checked:** KiCad's Electrical Rules Check on the schematic and Design Rules Check on the board find nothing, and the board matches its schematic pad for pad. `tools/check.mjs` also compares the schematic with the circuit of the drawings, connection by connection, so the two cannot drift apart.
- **Not checked:** no board has been made yet.

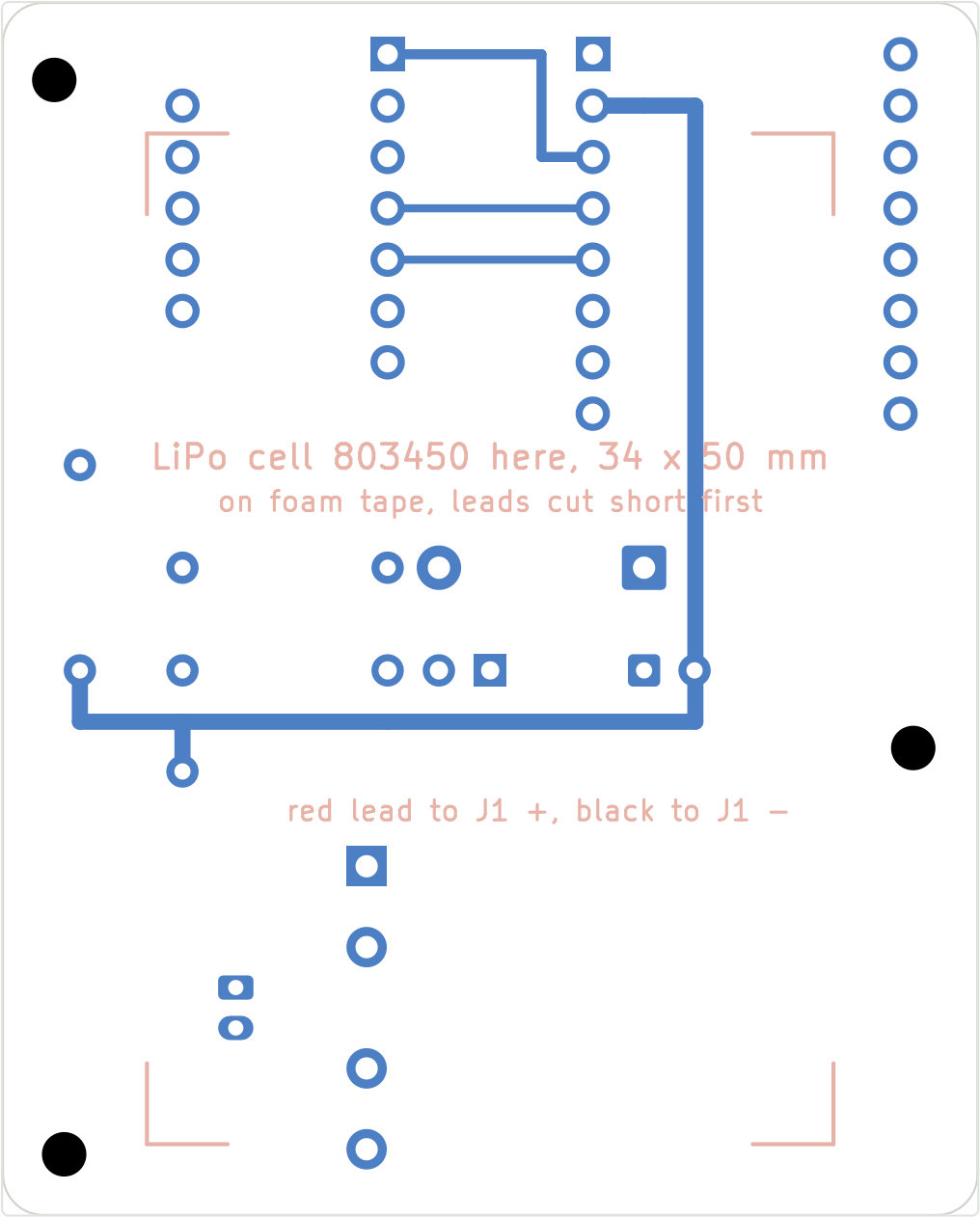
## One check before ordering

**The ESP32.** Its two rows of pins must be 15.2 mm apart, middle to middle. On a zero PCB that is 5 empty holes between a pin and the pin opposite it. Anything from 15.0 to 15.5 mm fits.

The charger module needs no check: it is joined with wires, so any size fits its marked place, up to 28 by 18 mm.

## The two sides






The schematic is in diagrams/pcb-schematic.svg, and on the last page of `diagrams/diagrams.pdf`.

## Order it

1. Go to any maker that takes Gerber files, for example JLCPCB or PCBWay, and upload `pcb/sswingg-bat-gerbers.zip` as it is. Do not unpack it.
2. Choose these settings. Leave everything else as it is.

| Setting | Choose |
|---|---|
| Layers | 2 |
| Thickness | 1.6 mm. 1.0 mm is lighter on the bat and also works. |
| Material | FR-4 |
| Surface finish | HASL lead free |
| Copper | 1 oz |

| Setting | Choose |
|---|---|
| Quantity | the smallest they offer, usually 5 |
| Assembly | none. Every part goes through holes and is soldered by hand. |

3. Before paying, open the maker's preview. It must look like the two drawings above, with `USB` at the top edge, the charger's `USB–C` mark at the left edge, and the cell's corner marks on the back.

## Parts

pcb/bom.csv is the list KiCad makes from the schematic, with the name to search for in a shop. The parts are the ones of the shopping list, minus the zero PCB and the wire, plus one JST socket, foam tape and four short pieces of wire.

| Ref | Part | Notes |
|---|---|---|
| U1 | ESP32-C3 Super Mini | In two 8-pin female headers. |
| U2 | GY-BMI160 | In a 7-pin female header for its pins, and a 5-pin female header for its second row, which only holds it level and is joined to nothing. |
| D1 | 1N5817 diode | **The stripe goes on the square pad, marked `K`.** |
| SW1 | SS12D00G3 slide switch | Either way round. |
| C2 | 220 µF electrolytic | **The long leg (+) goes in the square pad**, next to the `+` mark. |
| R1, R2 | 100 kilohm | Only for the battery reading of stage 7. Leave the holes empty otherwise. |
| C1 | 100 nF | Only for stage 7, like R1 and R2. |
| U3 | TP4056 charger module with protection | On foam tape in the dashed outline, USB-C at the left edge. Four short wires join its `OUT+`, `B+`, `B–` and `OUT–` to the pads with the same names. Its `IN+` and `IN–` stay free. |
| BT1 | The LiPo cell, 803450 | On the back, inside the four corner marks, on foam tape. Its plug goes round the edge into J1. |
| J1 | JST PH 2.0 mm socket, 2 pins, vertical (B2B-PH-K) | The cell plugs in here. If your cell has another plug, cut it off and solder the red lead into the hole marked `+`, the black into `–`. |
| H1 to H3 | Mounting holes, 2.2 mm | For M2 screws or a cable tie, or use foam tape under the board. |

**Check the cell's plug before plugging it in.** Makers wire JST plugs both ways round. With the multimeter on DC volts, touch the red probe to the `+` hole of J1 from the underside while the plug is in, and the black to `–`: it must read plus 3.7 to 4.2 volts. A minus reading means the plug is wired the other way: swap the two wires in the plug, or cut it off and solder the leads in. Plugged in the wrong way round, the charger is destroyed.

## Solder it

Lowest parts first, so that the board lies flat on the table while you solder.

1. **D1, R1 and R2, C1.** Bend the leads, push them through, solder on the underside, cut the leads short. Check the stripe of D1 first.
2. **SW1 and C2.** Check the long leg of C2 first.
3. **The four female headers**: two 8-pin for the ESP32, a 7-pin and a 5-pin for the sensor. Plug the ESP32 and the sensor into them before soldering, so the pins stay straight and the two sensor rows line up. The ESP32's USB socket goes to the edge marked `USB`.
4. **J1**, the battery socket.
5. **The charger module.** Solder a short piece of wire, about 3 cm, into each of its `OUT+`, `B+`, `B−` and `OUT−` holes. Stick the module in the dashed outline with foam tape, USB-C socket flush with the left edge. Solder each wire into the pad with the same name beside it: `OUT+` to `OUT+`, and so on. Check each one twice: a swapped `B+` and `OUT+` stops the protection working.
6. **The cell, last.** Cut every lead on the back of the board short, as close to the board as you can. Cover the back with foam tape, 1 to 2 mm thick, so that no lead can press into the cell. Stick the cell on it, inside the four corner marks, and take its plug round the edge into J1. A LiPo cell pierced by a lead can catch fire.

**No diode yet?** Solder a piece of wire through D1's two holes. Then switch the battery off before plugging a USB cable into the ESP32, every time, as in the zero PCB guide.

## Check it before any power

With the multimeter on continuity, the setting that beeps, and nothing plugged in:

- It must **not** beep between the `+` and `−` holes of J1, or between the charger's `OUT+` and `OUT−`.
- It must **not** beep between the ESP32's `5V` and `G` pins, or between `3V3` and `G`. The labels are printed next to its socket.
- It **must** beep between `G` of the ESP32 and `GND` of the sensor, `3` and `SDA`, and `4` and `SCL`.

Then follow stage 6 for the first power on the battery.

## Change the board

Open `pcb/sswingg-bat.kicad_pro` in KiCad 10. Change the schematic first, then in the board editor use **Tools > Update PCB from Schematic**, and move or route what changed. The tracks for 5V, GND and the battery lines are 0.8 mm wide, the signal tracks 0.4 mm. Keep them so.

When it looks right, run:

```
node tools/make_pcb_files.mjs
node tools/check.mjs
```

The first one runs KiCad's checks and stops if anything is wrong. Only then does it make the Gerber zip, the parts list and the drawings again. The second one confirms that the schematic is still the circuit of the drawings. If you change the circuit on purpose, change `NETS` in `diagrams/make_diagrams.py` too.